

# A path planning algorithm for mobile robot based on water flow potential field method and beetle antennae search algorithm<sup>☆</sup>

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## ABSTRACT

Aiming at the problems that beetle antennae search algorithm is difficult and easy to avoid obstacles when solving path planning problems, and falls into local optimization, which leads to low efficiency, a path planning algorithm of mobile robot combining the water flow potential field method and the beetle antennae search is proposed. Path planning divides the global path into segments by setting up segmented sites by using the beetle genetic operator. Local path planning between sites is classified according to the characteristics of obstacles, and the artificial potential field method is used to guide the search. The nature of water flow method is used to plan the obstacle avoidance route, optimize the obstacle avoidance process, and effectively avoid falling into local trap obstacles. Finally, the site coordinates are optimized by the beetle antennae search algorithm to improve the path quality and prevent the path from falling into local optimum. The simulation results show that the algorithm can effectively avoid obstacles in various path planning environments, has the characteristics of short time consumption, good optimization effect, and is not easy to fall into local optimization, and is an efficient algorithm for solving the path planning problem of mobile robots.

## 1. Introduction

At present, the development of science and technology is becoming more and more intelligent [1], and the automatic driving in intelligent transportation [2] is influenced by the calculation of deep learning [3–5], and its ability to detect the environment is very excellent, and an excellent path planning algorithm that can plan the path of the obtained environmental information has become particularly important.

Path planning is a hot issue in the research field of mobile robots. The purpose of path planning is to find a connected and barrier-free optimal path between two points in an environment with obstacles. An excellent path planning algorithm can greatly improve the performance of mobile robots.

The methods of solving robot path planning can be divided into two categories: traditional algorithm and intelligent algorithm. Many scholars at home and abroad have made in-depth research on path planning and made many achievements. In recent years, with the rise of artificial intelligence, more and more intelligent algorithms have been applied to the path planning optimization of mobile robots, such as particle swarm optimization, genetic algorithm, ant colony algorithm [6–8], etc. Genetic algorithm has

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strong adaptability in various fields, but its optimization effect in path planning is not particularly outstanding. The foraging process of ant colony algorithm is highly compatible with path planning. Ant colony algorithm is also widely used in robot path planning, but it also has the problems of slow convergence speed and low search efficiency.

The beetle antennae search algorithm (BAS) proposed in recent years is suitable for dealing with multidimensional problems, which has strong directionality, simple code and easy implementation. It is a good choice to use the beetle antennae search algorithm to optimize the path planning. In the literature [9], the author uses the beetle as the robot itself for path planning operation, and generates an effective path through the searching property of the beetle antennae, but it is often difficult to find an excellent path, and when faced with some trap obstacles, it is easy to fall into the situation that the local optimum cannot be solved. Therefore, in many studies, the search algorithm of beetle antennae is combined with other algorithms to optimize the path planning problem. ZHOU and ZHANG put forward a new intelligent method combining the search mechanism of beetle antennae search algorithm with particle swarm optimization algorithm in literature [10] and literature [11], respectively, and applied it to path planning. In literature [12], ZHANG combined the search algorithm of beetle antennae with ant colony algorithm to optimize the ant's travel rule, and literature [13]. However, when BAS algorithm is applied to path planning as a pathfinder, it is often affected by map obstacles and other factors, and it is easy to fall into local optimal solution, and it is difficult to give full play to the optimization ability of BAS algorithm in solving problems.

In the process of searching the path in path planning, it is very critical to find path and avoid obstacles effectively, especially to avoid obstacles. To solve the problems of obstacle avoidance and path finding, some people have applied it to path planning by improving artificial potential field method [14] and water flow method [15] respectively. Yang [16] set the speed increment to avoid obstacles by calculating the distance between obstacles and robots. Qi [17] put forward the idea of water flow method. To make the path safer and faster, Ding [18] obtained a smooth and safe optimized path by iteratively improving the gravitational function of the artificial potential field, and LIU [19] solved the problem that the artificial potential field method is easy to fall into local optimum by improving the repulsive force function of the artificial potential field. In the aspect of combination with intelligent algorithm, Li [20] and Dong [21] adopted ant colony algorithm combined with artificial potential field to improve the planning path. It can be found that the performance of artificial potential field method and water flow method is very suitable for dealing with obstacle avoidance in path planning, but the strategy needs to be optimized and improved.

So, In this paper, a path planning algorithm of mobile robot is proposed, which combines the water flow potential field method and the beetle antennae search algorithm. The path planning is segmented by designing the beetle operator, and the problem of solving path planning by the beetle antennae search algorithm is returned to the problem of parameter optimization, so that the optimization performance of the beetle antennae search algorithm is better exerted. In local path planning in segmented path planning, a new path planning strategy is designed by combining artificial potential field method and water flow method, which is used to deal with obstacles in path planning and ensure the safety and efficiency of path planning. Finally, the beetle antennae search algorithm is used to optimize the beetle operator and improve the quality of the path.

## 2. Artificial potential field method and water flow method

In order to solve the problems of obstacle avoidance and low efficiency in path planning, this paper designs a new effective strategy to solve the obstacle avoidance problem in path planning from the artificial potential field method and water flow method.

### 2.1. Artificial potential field method

Artificial potential field method [15] is a virtual force method proposed by Khatib in 1985. As its name implies, its basic idea is to artificially imitate a potential field model set up by the potential energy of nature, and to establish an artificial potential field model in a map for studying path planning. It is assumed that the target point produces a gravitational force on the robot, while the obstacle produces a repulsive force on the robot, which makes the robot search for an effective and feasible route to reach the target point under the combined action of gravitational force and repulsive force. When the robot is located at a certain point  $q$ , the potential energy is expressed by the potential function  $U(q)$ .

The potential function of a certain point  $q$ :

$$U(q) = U_{att}(q) + U_{rep}(q) \quad (1)$$

$U_{att}(q)$  represents the gravitational potential function of this point:

$$U_{att}(q) = \frac{1}{2} \varphi d^2(q, q_{goal}) \quad (2)$$

Among them,  $\varphi$  is the gravitational factor, which controls the size of the attractive force, and  $d$  represents the distance between the current point  $q$  and the target point  $q_{goal}$ .

$U_{rep}(q)$  represents the repulsive potential function of this point:

$$U_{rep}(q) = \begin{cases} \frac{1}{2} \gamma \left( \frac{1}{D(q)} - \frac{1}{Q^*} \right)^2, & D(q) \leq Q^*, \\ 0, & D(q) < Q^*. \end{cases} \quad (3)$$

In which,  $\gamma$  is the repulsive force factor, which controls the magnitude of repulsive force,  $D(q)$  represents the distance between point  $q$  and adjacent obstacles, and  $Q^*$  represents the critical value of the action distance of repulsive force of obstacles on robots.

The above is all the potential energy of the robot under the artificial potential field model. We can intuitively observe that the robot is moving forward by the gravity from the target point and avoiding obstacles by the repulsion of obstacles, so as to generate an effective path. However, there are still many defects under the combination of two forces, one of which is the lack of the ability to avoid concave obstacles. Once caught in a concave obstacle trap, it is difficult to jump out and find an effective path successfully.

## 2.2. Water flow method

Water flow method [16] is a new path planning algorithm that simulates water flow movement, which was proposed by Han Xuedong and others in 2004. By observing the movement principle of water in nature, we can find that there are two major movement rules of water flow: “water flows down” and “water over-flows when it is full”. That is:

- (1) Water flows from higher places to lower places
  - (2) Water can fill the space, and then the water overflows from both ends of the space and continues to flow to lower places.
- The effect of target position on water flow can be expressed as:

$$VH_g(x, y) = K_g \times D_g \quad (4)$$

Where  $K_g$  is a constant,  $D_g$  represents the distance from the target point to the water flow, and the influence of obstacles on the water flow can be expressed as:

$$VH_{obs}^i(x, y) = \begin{cases} K_{obs} - \rho D_r, & \rho D_r \leq K_{obs}, \\ 0, & \rho D_r > K_{obs}. \end{cases} \quad (5)$$

$$VH_{obs}(x, y) = \max\{VH_{obs}^i(x, y)\} \quad (6)$$

In which,  $K_{obs}$  is a constant,  $D_r$  represents the distance between obstacles and water flow, and  $\rho$  is the obstacle influence factor. During the whole route finding process, the influence of target position and obstacles on water flow can be expressed by the following equation:

$$VH(x, y) = \max\{VH_{obs}^i(x, y), VH_g(x, y)\} \quad (7)$$

The application of water flow method in path planning has strong target and robustness, and the general direction of water flow is determined by the current target point, that is, the nature of “water flows down”, which is actually very similar to the artificial potential field, but the obstacle avoidance strategy of water flow method is different, especially in the face of concave obstacle traps, the nature of water flow method “overflow when water is full” can well avoid the traps. Therefore, the strategy of combining water flow method with artificial potential field method in path planning can solve the problem of complex obstacle shape in obstacle avoidance.

## 2.3. Improved water flow potential field method(WFPFM)

By combining water flow method and artificial potential field method, water flow potential field method is designed. The water flow potential field method relies on the gravity provided by the potential field points to advance and the characteristics of water flow to plan a safe and efficient obstacle avoidance route. This method solves the problem of insufficient obstacle avoidance ability of artificial potential field method in complex environment. The new strategy can deal with different complex environmental obstacles well.

The improved flow potential field method has different ways to deal with different obstacles in path planning, and different strategies are adopted by analyzing the types of obstacles.

When the water flow does not perceive obstacles, it moves forward under the influence of the gravity of the target point in the environment. The velocity of water flow under the condition that it is only attracted by the gravity of the target point is expressed as:

$$V = V_x + V_y \quad (8)$$

In which,  $V_x$  and  $V_y$  represent the velocity of water flow on  $x$  axis and  $y$  axis respectively:

$$V_x = \rho_x(D) \quad (9)$$

$$V_y = \rho_y(D) \quad (10)$$

$D$  indicates the distance from the target point( $X, Y$ ) to the water flow ( $x, y$ ):

$$D = \sqrt{(X - x)^2 + (Y - y)^2} \quad (11)$$

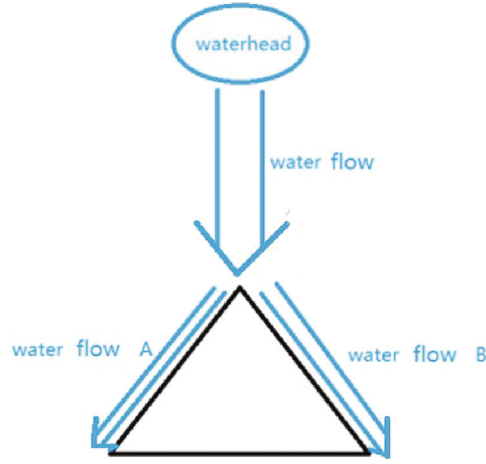


Fig. 1. Water flow when encountering a convex smooth obstacle.

Therefore, the position updating equation of water flow is expressed as:

$$C_{next} = C_{now} + M(V) \quad (12)$$

In the above formula (12),  $a$  represents the distance that the water flow is affected by gravity.

Then analyze the situation when obstacles are detected. In the process of searching the path by the algorithm, we generate a special detection water flow at the water source. The detection water flow has an inseparable nature. When obstacles are encountered, different information will be fed back according to different obstacles, and different processing methods will be made.

#### Class I: Convex smooth obstacles that do not affect the flow

As shown in the figure, when the water flow meets the obstacle shown in Fig. 1, the speed of the water flow changes due to the repulsive force of the obstacle, but it does not affect the water flow's continued progress. It is only necessary to update the speed and position of water flow according to the repulsive force of obstacles, and then continue to search for the path to the target point.

The update of water flow velocity is expressed as:

$$V_{next} = V_{now} + v(f_{obs}) \quad (13)$$

In Eq. (13), the magnitude  $v(f_{obs})$  of the obstacle resistance is equal to the current velocity of the water flow perpendicular to the obstacle plane, so that the water flow moves along the obstacle.

The update of water flow position is expressed as:

$$C_{next} = C_{now} + M(V_{next}) \quad (14)$$

In the above formula (14),  $a$  represents the distance that the water flows along the obstacle.

#### Class II: Trap type obstacles affecting water flow

As shown in the figure, when the obstacle encountered by the water flow is a trap obstacle like Fig. 2, according to the characteristic of “overflow when water is full” of the water flow method itself, the water flow will accumulate in the groove, and then overflow and flow to both sides. It can be seen from the figure that a lot of water is gathered in the groove, which falls into a local trap for a long time in the route planning, and then escapes from both ends.

Faced with this problem, the improved water flow potential field method detects such obstacles by detecting water flow, then scans the obstacles, and finds the maximum value in the two-dimensional function, as shown in Fig. 3:

The two overflow points in the figure satisfy the following properties in the coordinate system:

$$f'(x) = 0 \quad (15)$$

$$f'(x + \varepsilon) = 0 \quad (16)$$

$$f'(x - \varepsilon) = 0 \quad (17)$$

In mathematical expression, overflow point is the extreme point in the graph. Find the extreme point in the graph by Eq. (15),(16),(17). Calculate the distances  $D_1$  and  $D_2$  between extreme point and water flow respectively. According to Eqs. (18),(19), a new water flow velocity is generated based on  $D_1$  and  $D_2$ . At the same time, the water flow is only affected by the gravity of a

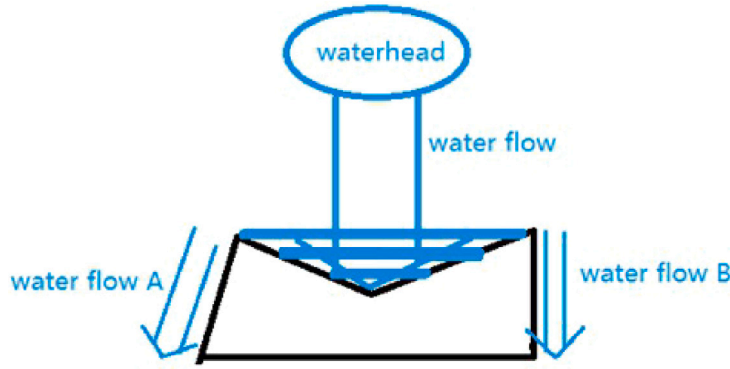


Fig. 2. Water flow when encountering a trap obstacle.

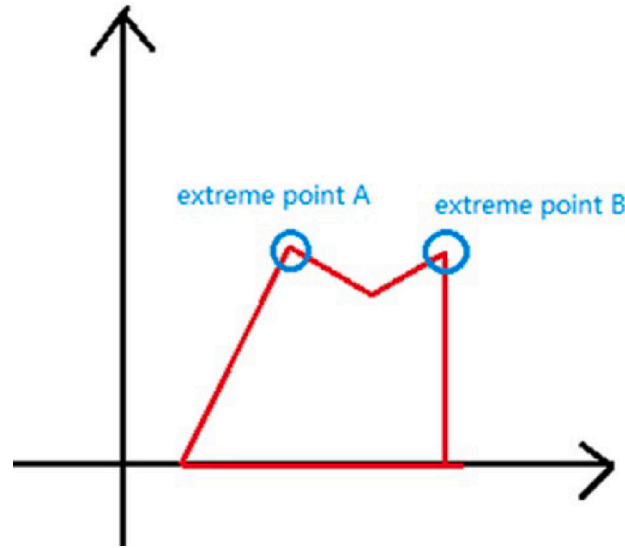


Fig. 3. Schematic diagram of obstacle overflow point (extreme point).

target point, and usually two extreme point are obtained, so the speed of the water flow is updated by random selection. As Eq. (20).

$$V_1 = \rho(D_1) \quad (18)$$

$$V_2 = \rho(D_2) \quad (19)$$

$$V = \begin{cases} V_1, rand \leq 0.5, \\ V_2, rand > 0.5. \end{cases} \quad (20)$$

After changing the speed of water flow by updating the target point, the obstacle avoidance effect diagram is achieved :Fig. 4  
The specific implementation steps are as follows:

- (1) Initial water source position, initial terminal gravitational potential field position, initial water flow velocity.
- (2) A probe water flow released by the water source is moved to the current potential field point by the potential field force.
- (3) When encountering concave obstacles, detect the obstacles, find two adaptive obstacle avoidance points, and jump to step 4.
- (4) When a convex obstacle is encountered, update the water flow speed according to the position of the obstacle and the repulsion of the obstacle to the water output, and jump to step 2.
- (5) When the water flow reaches the potential field point, a path is generated from the current water source to the potential field point. If there are other potential field points to be passed, update the current water source point as the current potential field point, update the current potential field point as the next adjacent potential field point, and jump to step 2; if not, jump to step 5.

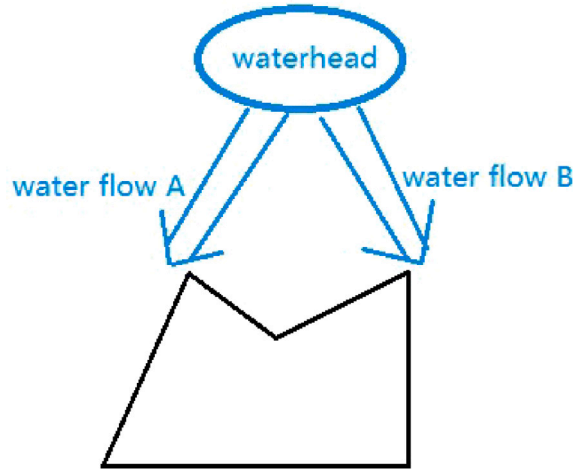


Fig. 4. Schematic diagram of improved water flow direction.

- (6) Randomly select an obstacle avoidance point to update as the current potential field point, and the potential field point releases gravity to the water flow to update the water flow speed. Jump to step 2.
- (7) Output the path from the initial water source to the end point, and end the path finding.

Through the above strategy, the improved current potential field method has the ability to deal with all kinds of obstacles in path planning, and can be well applied to path planning.

### 3. Improved water flow potential field method incorporating search algorithm for beetle antennae (WPFBAS)

#### 3.1. Beetle antennae search algorithm

Beetle Antennae Search Algorithm (BAS) [22] is a new intelligent optimization algorithm proposed by jiang in 2017, which is realized by imitating the foraging activity law of beetles. Compared with some swarm optimization algorithms, the beetle antennae search algorithm only needs one beetle to complete the optimization, which greatly reduces the calculation cost.

Although beetle is a moving creature in three-dimensional space, the beetle Antennae Search Algorithm can optimize any dimensional function problem, which is more like a derivative of the simulation of the activity of beetle. The beetles are placed in a multi-dimensional space, and the orientation of the beetles is random. Then, according to the fitness values detected by the left and right tentacles, the better side is selected to move, and the optimized solution is obtained through multiple iterations. The implementation of the search algorithm for beetles is as follows.

The random direction is expressed as:

$$Rand = rand(n) \quad (21)$$

$$dir = \frac{Rand}{norm(Rand)} \quad (22)$$

In the above equation,  $n$  represents the dimension of the parameter space,  $rand(n)$  represents random value taking in the  $n$  dimension, and  $dir$  is a random vector, representing the orientation of the beetle.

Position of left and right tentacles of beetles:

$$X_{left} = X - \eta \times dir \quad (23)$$

$$X_{right} = X + \eta \times dir \quad (24)$$

In Eqs. (23) and (24), it is a constant, indicating the current coordinates of the beetle.

Update the location of beetle:

$$X_{next} = \max(f(X_{left}), f(X_{right})) \times step \quad (25)$$

As shown in the above equation, by comparing the fitness of the left and right tentacles, the better position is selected as the new coordinates of the beetle, and the process of searching for the target of the beetle in space is completed by iterative updating. The following is a conceptual diagram of the BAS algorithm in the two-dimensional case and a search diagram for the optimal function solution in the two-dimensional environment: Fig. 5, Fig. 6.

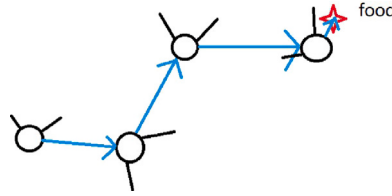


Fig. 5. Beetle antennae search food.

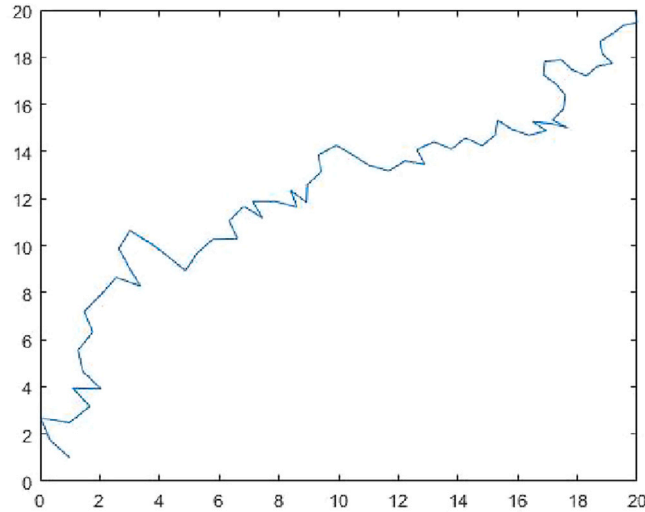


Fig. 6. Search path of BAS.

It can be seen that BAS algorithm has strong optimization and high efficiency. However, in the research of path planning, it is found that BAS algorithm lacks obstacle avoidance and is not suitable for path planning search as an individual in an obstacle environment. In order to give full play to the ability of BAS algorithm to optimize and solve multidimensional problems, a strategy is designed to segment the whole path planning process, take the segmentation points as parameters and use BAS algorithm for optimization.

### 3.2. Water flow potential field method integrated with beetle antennae search algorithm (WPFBAS)

In the design of the final path planning algorithm WPFBAS, the excellent performance of BAS algorithm in solving multidimensional problems is used. In the overall view, the stations are set up in the path planning to plan the path in sections, and the connection mode between stations adopts the strategy of water flow potential field method to plan local paths. The excellent performance of the water flow potential field method ensures the safety and efficiency of the local path planning. BAS algorithm is used to optimize the coordinates of each station, so as to optimize the path to ensure the Excellence of the overall path.

The mathematical model is as follows:

As shown in Fig. 7, in order to segment the map better, a certain amount of average segmentation is performed on the map, and then the site coordinates obtained after segmentation are set as the beetle genetic operator with reference to the strategy of the genetic operator in the genetic algorithm, as shown in the following equation.

$$X = [X_1, X_2, X_3, \dots] \quad (26)$$

In Eq. (26)  $X$  represents the set of all coordinates.

The connection between stations is searched for local path planning by the water flow potential field method, as Eq. (27):

$$Path(X_n, X_{n+1}) = WPFPM(X_n, X_{n+1}) \quad (27)$$

$$Path(X) = Path(X_1, X_2, X_3, \dots) \quad (28)$$

Eq. (28) the path connecting the sites constitutes the whole path of path planning. As shown in Fig. 8.

According to the optimization strategy of the beetle antennae search algorithm in the previous article, iterative optimization of the beetle genetic operator representing site coordinates.

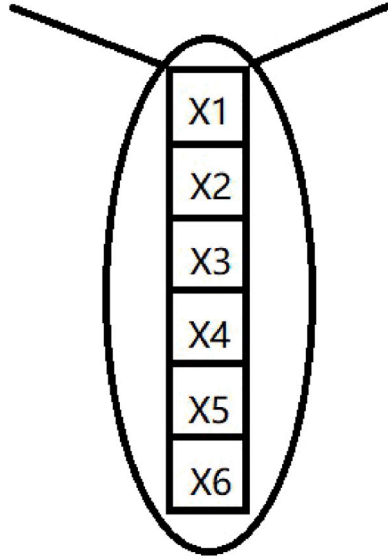


Fig. 7. Path beetles.

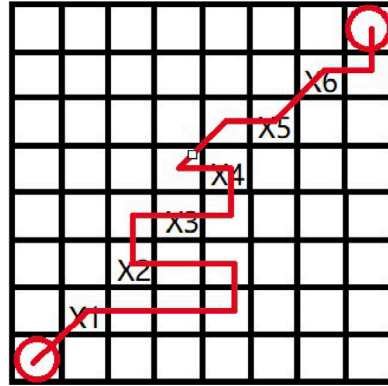


Fig. 8. Path map.

Use Eq. (22), (23) to randomly generate the optimization direction:  $dir$

Bring in the coordinate set  $X$  to generate the coordinate set of the left and right antennae of beetles:  $X_{left}$ ,  $X_{right}$ . In Eqs. (29) and (30),  $step$  represents the step size of the algorithm.

$$X_{left} = X - \eta \times dir \times step \quad (29)$$

$$X_{right} = X + \eta \times dir \times step \quad (30)$$

Eqs. (31) and (32) represent the corresponding total paths generated by  $X_{left}$  and  $X_{right}$  through  $WPFM$  connection sites according to Eqs. (27) and (28). Eq. (33) select a better path to update the genetic operator of beetles (site coordinate set):

$$Path_{left} = WPFM(X_{left}) \quad (31)$$

$$Path_{right} = WPFM(X_{right}) \quad (32)$$

$$X_{next} = \begin{cases} X_{left}, Path(X_{left}) \leq Path(X_{right}) \\ X_{right}, Path(X_{left}) > Path(X_{right}) \end{cases} \quad (33)$$

Through Eq. (34), (35), the elite strategy is adopted to update and save the optimal path and site set in the iterative process

$$BestPath = \begin{cases} Path(X), i = 1 \\ Path(X_{next}), X_{next}, i > 1 \text{ \& } Path(X_{next}) < BestPath \end{cases} \quad (34)$$



**Table 1**

In environment Fig. 10, simulation data comparison between WPFBAS and BAS and PFA.

| Environment Fig. 10 | BAS    | PFA    | WPFBAS |
|---------------------|--------|--------|--------|
| Optimal             | 36.142 | 34.582 | 28.627 |
| Average length      | 43.897 | 35.611 | 28.627 |
| Average Time        | 0.034  | 0.207  | 0.262  |

$$BestX = \begin{cases} X, & i = 1 \\ X_{next}, & i > 1 \text{ } \zeta \text{ } Path(X_{next}) < BestPath \end{cases} \quad (35)$$

Through the above algorithm flow, the path is iteratively optimized, and finally the optimal site set and path are returned.

In solving the robot path planning, the whole path planning and path optimization are carried out by using the beetle antennae search algorithm to segment the map, and the local path planning search is carried out by using the water flow potential field method for the segmented paths between stations.

In this way, the path planning algorithm is designed to solve the path planning problem, which not only avoids the defect of low efficiency and easy to fall into local optimization by directly using the beetle antennae search algorithm, but also gives full play to the excellent performance of the beetle antennae search algorithm in solving multi-dimensional optimization problems.

#### 4. Simulation experiment

In order to verify the effectiveness of the algorithm, this paper chooses to compare with different algorithms and select excellent improved ant colony algorithm for further performance comparison to verify the performance of this algorithm.

The simulation experiments are all carried out in a static environment. Assuming that the robot has completely detected the map information through the sensor, the path planning is carried out on this basis.

The performance of this algorithm is mainly affected by the parameters in BAS algorithm. The antenna spacing of beetle in formula (23) is  $\eta$ , and the optimization step size of BAS algorithm in formula (25) is  $step$ . The parameter settings of  $\eta$  and  $step$  jointly affect the perception interval of beetles in BAS algorithm. If the parameter settings are too large, the algorithm will be difficult to converge, and if they are too small, the randomness of the algorithm will be insufficient to jump out of the local optimal solution.

Debugging the parameters through experiments., the experimental parameters of the current algorithm have stable and excellent performance after manual adjustment, and can be used in simulation experiments to compare with other algorithms to verify the performance of the algorithm.

##### 4.1. Compared with beetle antennae search algorithm (BAS) and artificial potential field (PFA).

In order to prove the effectiveness and practicability of the improved algorithm, the performance of the algorithm is verified by simulation and comparison experiments with BAS algorithm and PFA algorithm. The algorithm runs in the same environment, and the simulation experiments are carried out for simple maps and complex maps respectively to verify the improved algorithm's efficient obstacle avoidance ability. As shown in Fig. 9, the implementation represents BAS, the dotted line represents PFA, and the dotted line represents WPFBAS.in the complex map with trap obstacles, both BAS algorithm and PFA algorithm cannot escape because of their own defects, but it can be clearly seen that the algorithm in this paper is not affected by complex map obstacles, and can avoid obstacles quickly and safely and complete path planning.

As shown in Fig. 10, BAS, PFA and WPFBAS can all complete the path planning in a relatively normal map, and the path planning results of ten simulation experiments are shown in Table 1. It can be seen that WPFBAS not only has excellent path results, but also can find the optimal path in ten simulation experiments.

##### 4.2. Compared with genetic algorithm (GA) and particle swarm optimization algorithm (PSO).

Because this paper is a fusion algorithm optimized by intelligent algorithm, two intelligent optimization algorithms, Genetic Algorithm and Particle Swarm Optimization, are selected for simulation test comparison to verify the superiority of WPFBAS.

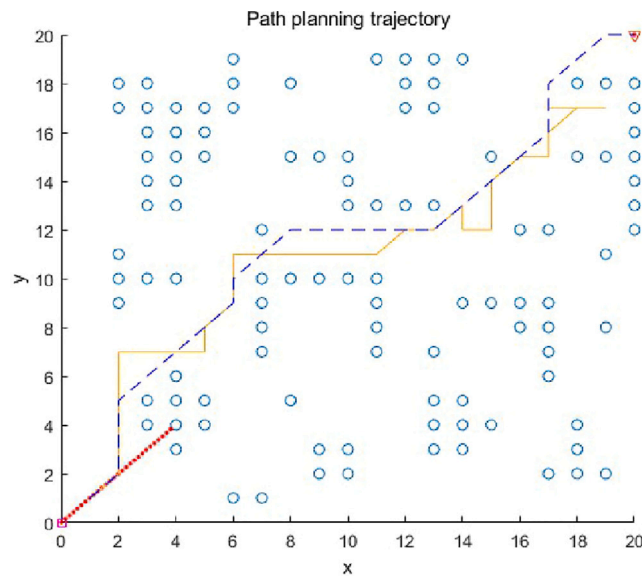
Fig. 11 and Fig. 12 respectively show the convergence trend of the paths of the algorithm and genetic algorithm in the environment of Fig. 11. It can be clearly found that the convergence speed of the algorithm is faster and the path is shorter. By observing Table 2, it can be found that the time consumption of the algorithm is shorter and the probability of finding the optimal path is higher, which is more excellent than the genetic algorithm.

It can be seen from the simulation test data in Table 2 that under the environment of Fig. 11, the optimal path found by WPFBAS is 30.3848, which is better than the optimal path found by genetic algorithm (30.9706), and the average time spent by WPFBAS is 0.187702, which is obviously better than the average time spent by genetic algorithm (0.501126).

Figs. 13 and 14 show the comparison between the WPFBAS and particle swarm optimization. It is also obvious that the WPFBAS is better than particle swarm optimization in path optimization. Table 3 also shows that the WPFBAS is more efficient and performs better, which proves the excellent performance of the algorithm.

**Table 2**  
Comparison of ten simulation data between WPFBAS and GA.

| ID | WPFBAS  |         |            | GA      |         |           |
|----|---------|---------|------------|---------|---------|-----------|
|    | Length  | Time    | Iterations | Length  | Time    | Iteration |
| 1  | 30.9706 | 0.15465 | 33         | 31.5563 | 0.56759 | 35        |
| 2  | 30.3848 | 0.12876 | 22         | 31.799  | 0.57714 | 34        |
| 3  | 30.3848 | 0.08287 | 20         | 32.9706 | 0.54598 | 11        |
| 4  | 30.3848 | 0.16794 | 16         | 31.799  | 0.47605 | 27        |
| 5  | 30.9706 | 0.09117 | 23         | 33.5563 | 0.39597 | 37        |
| 6  | 30.3848 | 0.08905 | 24         | 31.5563 | 0.58262 | 26        |
| 7  | 30.9706 | 0.31251 | 26         | 30.9706 | 0.41961 | 34        |
| 8  | 30.3848 | 0.30722 | 46         | 31.5563 | 0.41414 | 35        |
| 9  | 30.3848 | 0.09279 | 17         | 32.9706 | 0.40329 | 28        |
| 10 | 30.3848 | 0.27004 | 19         | 31.5563 | 0.62887 | 29        |



**Fig. 9.** Complex map path comparison diagram.

**Table 3**  
Comparison of ten simulation data between WPFBAS and PSO.

| ID | WPFBAS  |         |            | PSO     |        |           |
|----|---------|---------|------------|---------|--------|-----------|
|    | Length  | Time    | Iterations | Length  | Time   | Iteration |
| 1  | 28.6274 | 0.15465 | 25         | 31.4558 | 3.7886 | 17        |
| 2  | 28.6274 | 0.3023  | 5          | 30.6274 | 3.5709 | 33        |
| 3  | 28.6274 | 0.30876 | 31         | 30.0416 | 3.6735 | 1         |
| 4  | 28.6274 | 0.08287 | 20         | 30.0416 | 3.8203 | 46        |
| 5  | 28.6274 | 0.09117 | 23         | 29.4558 | 3.5982 | 33        |
| 6  | 29.2132 | 0.08905 | 24         | 30.0416 | 3.7259 | 40        |
| 7  | 28.6274 | 0.31251 | 26         | 30.6274 | 3.7168 | 37        |
| 8  | 28.6274 | 0.30722 | 6          | 30.6274 | 3.647  | 10        |
| 9  | 28.6274 | 0.09279 | 17         | 30.6274 | 3.6312 | 37        |
| 10 | 28.6274 | 0.27004 | 19         | 30.3848 | 3.6379 | 15        |

By observing the test data in Table 2, we can observe that in the environment of Fig. 11, the optimal paths found by our algorithm and particle swarm optimization algorithm are 28.6274 and 29.4558, respectively, and the average time consumption is 0.2434505 and 3.68103. It can be seen that our algorithm is more excellent in time and optimal path.

Compared with genetic algorithm and particle swarm optimization in two environments, the WPFBAS is more effective and stable than genetic algorithm and particle swarm optimization in both time-consuming and path optimization results, and the optimal path is successfully found with great probability, which further proves the superior performance of the WPFBAS.

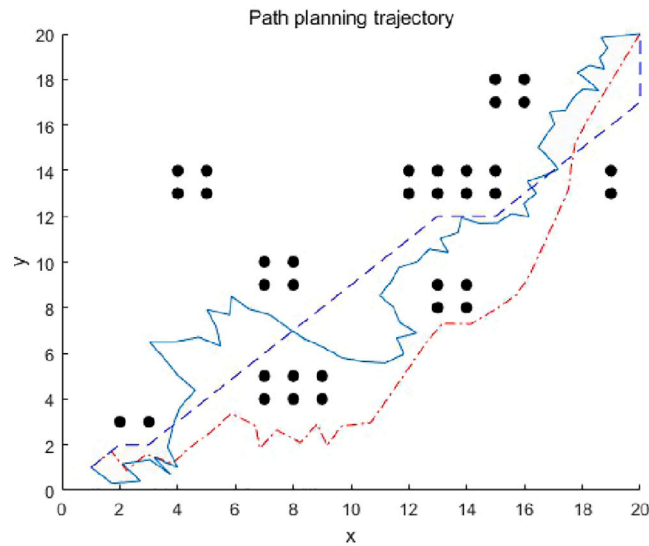


Fig. 10. Normal map path comparison diagram.

Table 4

In environment Fig. 15, simulation data comparison between WPFBAS and reference [18] and [19].

| Environment Fig. 15                  | Ref. [24] | Ref. [23] | WPFBAS  |
|--------------------------------------|-----------|-----------|---------|
| Optimal                              | 28.038    | 28.038    | 28.038  |
| Average                              | 32.894    | 31.301    | 28.1002 |
| Average iterations                   | 22        | 17        | 3       |
| Get the number of optimal solutions. | 2         | 7         | 9       |

Table 5

In environment Fig. 17, simulation data comparison between WPFBAS and reference [18] and [20].

| Environment Fig. 17                 | Ref. [25] | Ref. [23] | WPFBAS |
|-------------------------------------|-----------|-----------|--------|
| Optimal                             | 30.384    | 30.384    | 30.384 |
| Average                             | 31.898    | 31.898    | 31.024 |
| Average iterations                  | 24        | 16        | 2      |
| Get the number of optimal solutions | 6         | 8         | 10     |

#### 4.3. Compared with various improved ant colony algorithms

In this paper, through the simulation and comparison tests with various path planning algorithms in different aspects, it is found that the WPFBAS has excellent performance in path planning. Because ant colony algorithm is widely used to solve path planning problems, it is very representative, so this paper selects different improved ant colony algorithms for simulation and comparison tests to verify the performance of WPFBAS.

Fig. 15 shows the generation paths of the WPFBAS and the algorithms in Ref. [23] and Ref. [24] under environment Fig. 15. Since both of them can find the optimal path, we can find that the convergence trend in Fig. 16 shows that the WPFBAS converges faster. According to the simulation data in Table 4, it can be found that the WPFBAS takes less time, requires fewer iterations, and the average path found is more excellent.

Fig. 17 shows the paths generated by the WPFBAS and the algorithms in Ref. [23] and Ref. [25]. Respectively, since both of them can find the optimal path, we can find that the convergence trend of Fig. 18 shows that the WPFBAS converges faster. According to the simulation data in Table 5, it can be found that the WPFBAS takes less time, requires fewer iterations, and the average path found is more excellent.

Fig. 19 shows the generation paths of the WPFBAS and the algorithms in Ref. [26] and Ref. [27], the implementation represents Ref. [27], the dotted line represents Ref. [26], and the dotted line represents WPFBAS. According to the simulation data in Table 6, although both of them can find the optimal path, we can find that the WPFBAS takes less time and requires fewer iterations, and the average path found is more excellent.

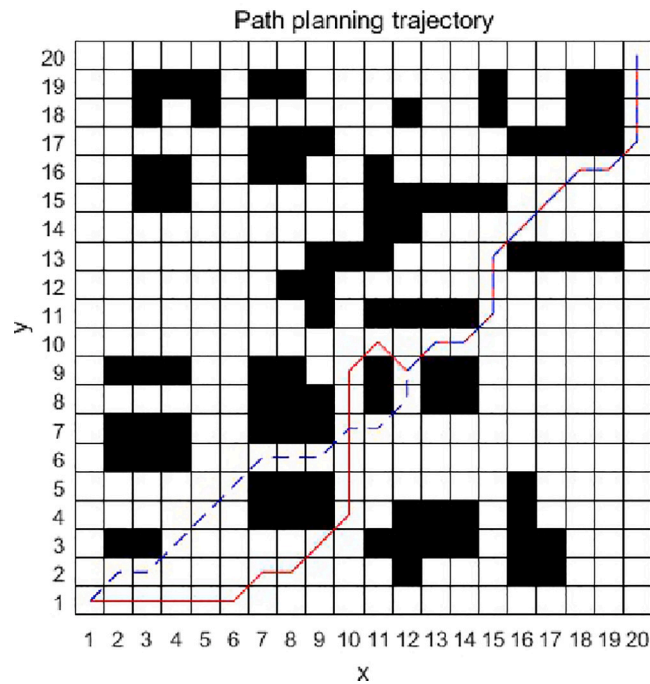


Fig. 11. Optimal path comparison diagram I.

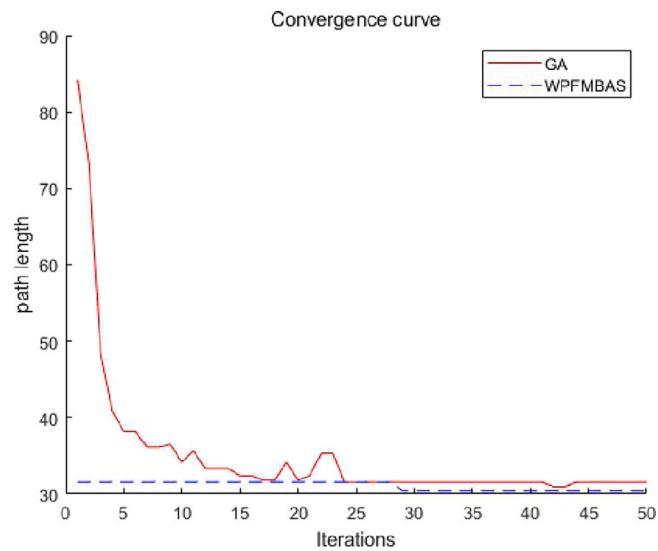


Fig. 12. Convergence comparison diagram I.

Table 6

In environment Fig. 19, simulation data comparison between WPFMBAS and reference 21 and 22.

| Environment Fig. 19 | Ref. [27] | Ref. [26] | WPFMBAS |
|---------------------|-----------|-----------|---------|
| Optimal             | 28.6274   | 28.6557   | 28.6274 |
| Average iterations  | 9         | 21        | 6       |
| Average time        | 0.34      | 0.63      | 0.1486  |

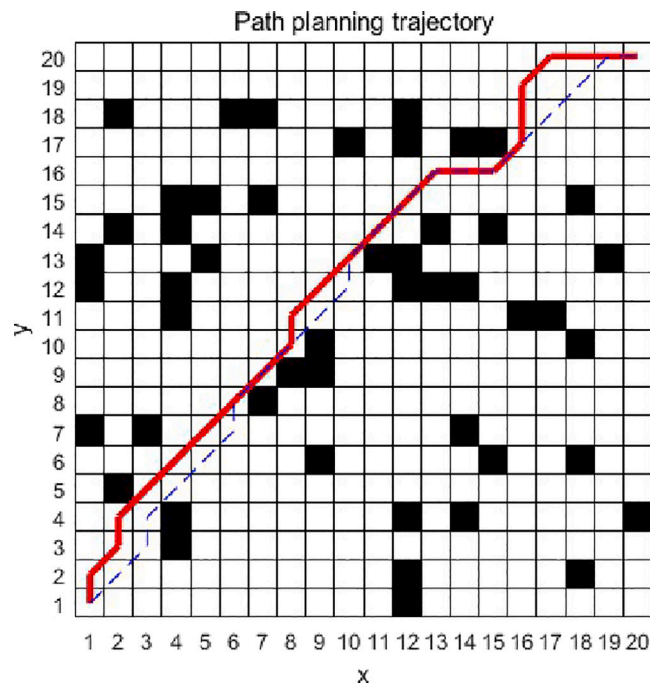


Fig. 13. Optimal path comparison diagram II.

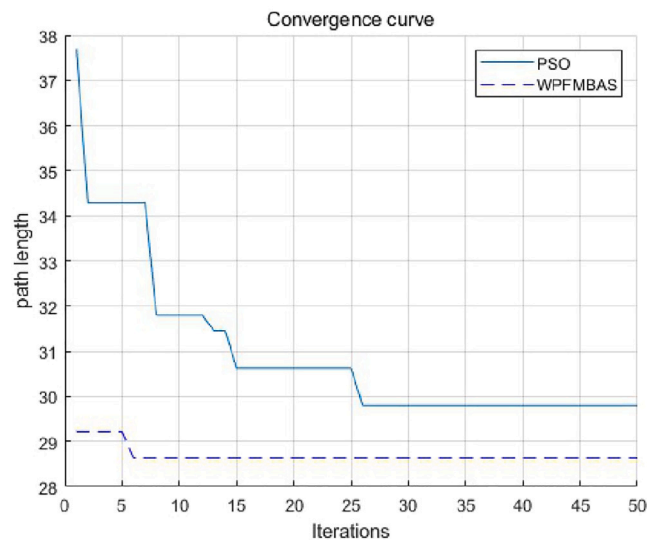


Fig. 14. Convergence comparison diagram II.

Fig. 20 shows the generation paths of the WPFMBAS and the algorithm in the Ref. [28], the implementation represents Ref. [28], the dotted line represents WPFMBAS, and both can find the optimal path. By continuing to observe the simulation data in Table 7, it can be found that the WPFMBAS takes less time and requires fewer iterations, and the average path found is more excellent.

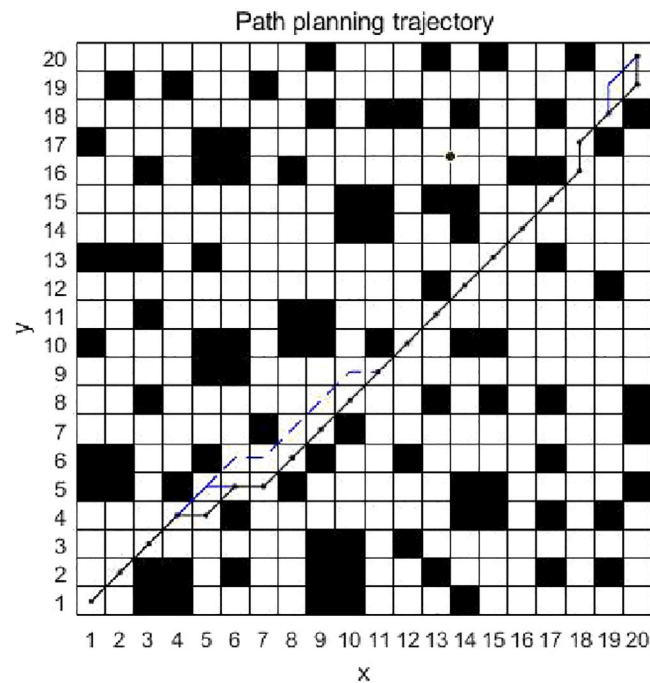
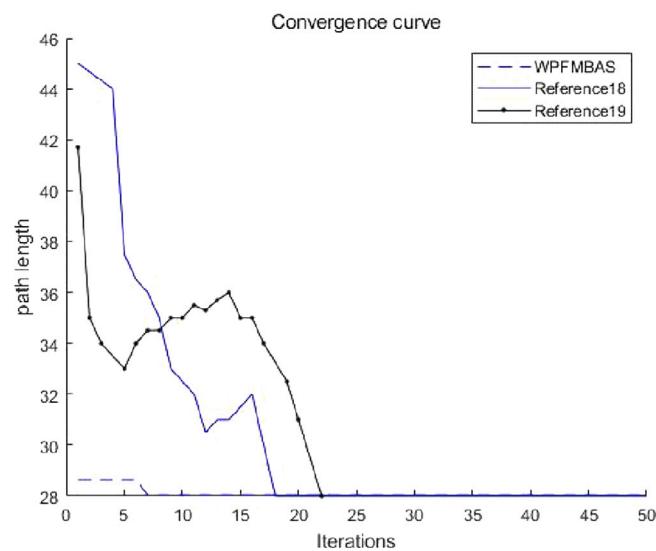
#### 4.4. Simulation summary

In different grid environments, compared with artificial potential field method, genetic algorithm and particle swarm optimization algorithm, and compared with various improved ant colony algorithms in different environments, it can be seen that WPFMBAS

**Table 7**

In environment Fig. 20, simulation data comparison between WPFMBAS and reference 23.

| Environment Fig. 20 | Ref. [28] | WPFMBAS |
|---------------------|-----------|---------|
| Optimal             | 30.69     | 28.6274 |
| Average iterations  | 16        | 1       |
| Average time        | 0.52      | 0.138   |

**Fig. 15.** Optimal path comparison diagram III.**Fig. 16.** Convergence comparison diagram III.

is faster, has better results and fewer iterations when dealing with the same problems, and it can cope with all kinds of obstacle environments, has strong applicability and can be well used for path planning and solving problems.

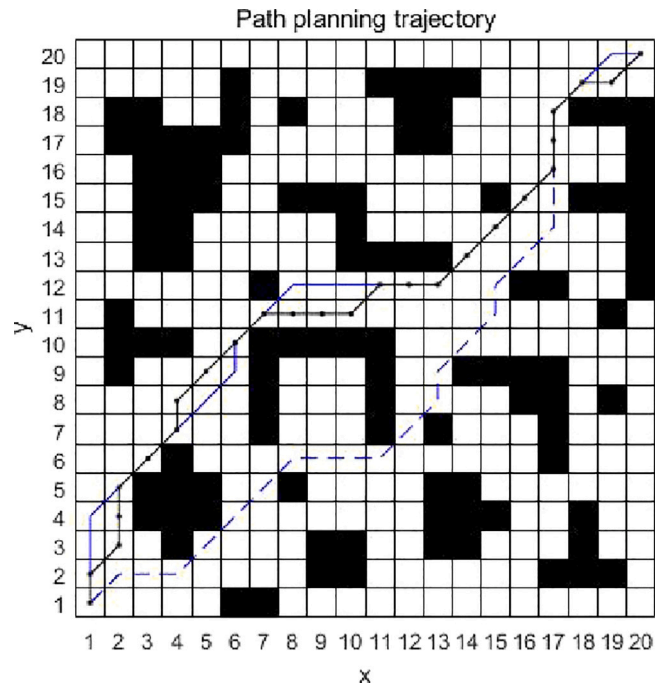


Fig. 17. Optimal path comparison diagram IV.

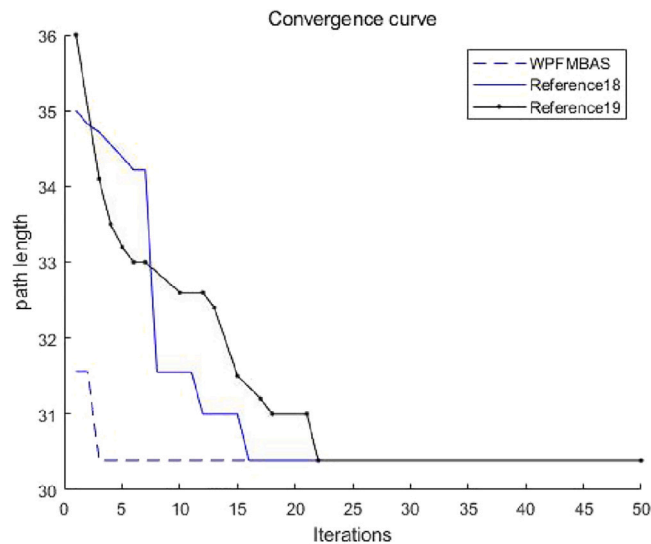


Fig. 18. Convergence comparison diagram IV.

## 5. Conclude

At present, intelligent algorithms have a good performance in various application fields. For the application field of path planning, there are many intelligent algorithms which often have many iterations, take a long time and have a general effect. Therefore, many scholars have adopted various improvement methods to optimize intelligent algorithms and improve their application effect in path planning.

Compared with other researchers' strategy of directly applying BAS algorithm to path planning, this paper adopts BAS algorithm to solve the problem of robot path planning from a macro perspective, and at the same time, in order to solve the problem of easily falling into local optimum and obstacle trap area in path planning, it is improved by combining water flow method and artificial

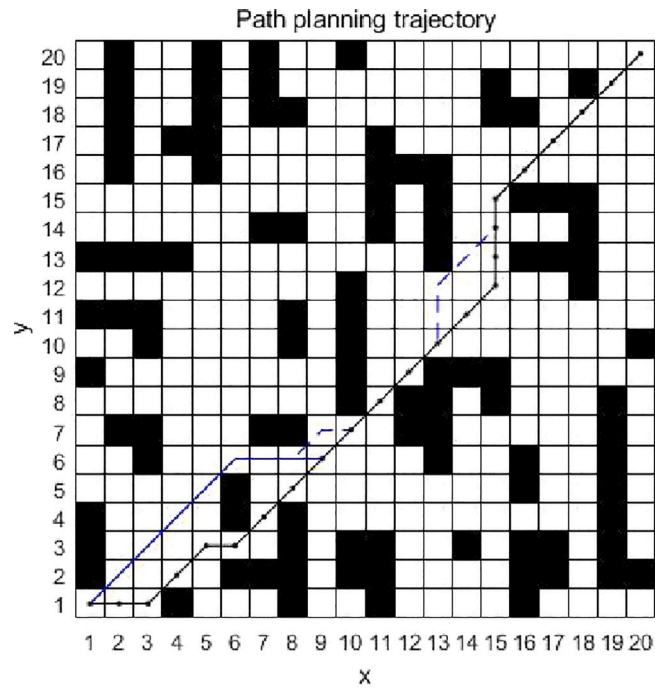


Fig. 19. Optimal path comparison diagram V.

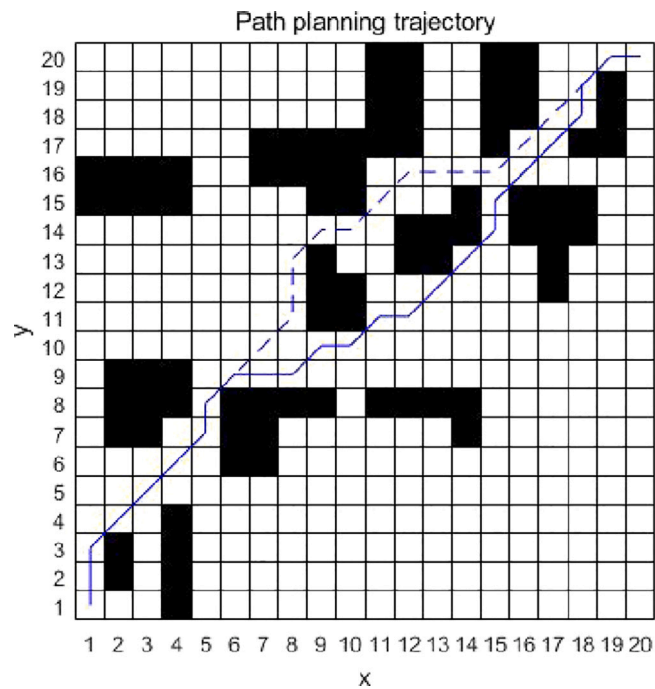


Fig. 20. Optimal path comparison diagram VI.

potential field method, so that the process of path finding has a strong direction, and obstacle avoidance operation is carried out for map environment. BAS algorithm gives the algorithm a broader global search ability and can effectively jump out of local optimum.

However, at present, this paper only studies the path planning for static obstacle avoidance in two-dimensional environment, and then plans to study the path planning and application in dynamic obstacle avoidance and three-dimensional environment.



## CRediT authorship contribution statement

**Zhiheng Yu:** Conceptualization, Methodology, Software, Writing – original draft, Formal analysis, Funding acquisition. **Jian Yuan:** Conceptualization, Methodology, Software, Writing – original draft, Formal analysis, Funding acquisition. **Yongsheng Li:** Data curation, Writing – review & editing. **Changan Yuan:** Resources, Investigation. **Song Deng:** Investigation, Project administration, Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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